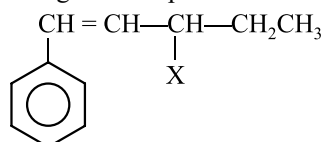


CHAPTER 7

Haloalkanes and Haloarenes

Classification of Organohalogen Compounds, Nature of C-X Bond

1. The given compound (2023)



is an example of _____.

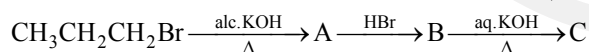
- vinyl halide
- benzylic halide
- aryl halide
- allylic halide

2. The correct sequence of bond enthalpy of 'C-X' bond is: (2021)

- $\text{CH}_3-\text{F} > \text{CH}_3-\text{Cl} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$
- $\text{CH}_3-\text{F} < \text{CH}_3-\text{Cl} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$
- $\text{CH}_3-\text{Cl} > \text{CH}_3-\text{F} > \text{CH}_3-\text{Br} > \text{CH}_3-\text{I}$
- $\text{CH}_3-\text{F} < \text{CH}_3-\text{Cl} < \text{CH}_3-\text{Br} < \text{CH}_3-\text{I}$

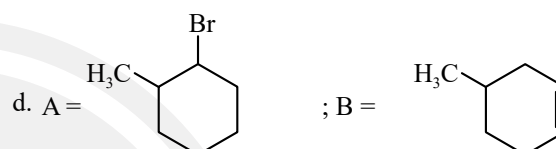
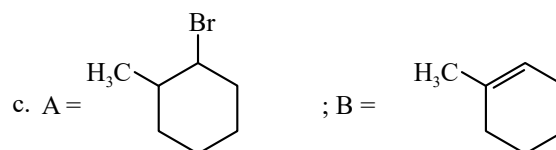
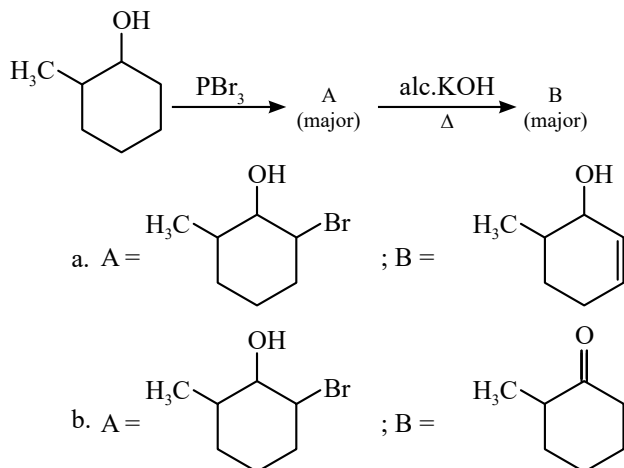
Haloalkanes and Chemical Reactions of Haloalkanes

3. The major product C in the below mentioned reaction is: (2024 Re)

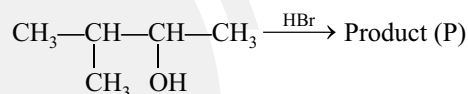


- Propan-1-ol
- Propan-2-ol
- Propane
- Propyne

4. Major products A and B formed in the following reaction sequence, are (2024)



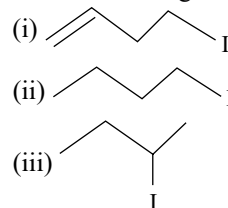
5. Consider the following reaction and identify the product (P). (2023)



3-Methylbutan-2-ol

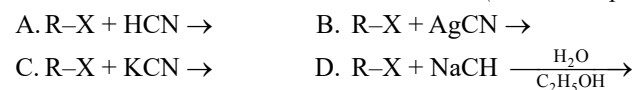
-
-
-
-

6. The correct order for the rate of α , β -dehydrohalogenation for the following compounds is (2023-Manipur)



- (i) < (ii) < (iii)
- (ii) < (i) < (iii)
- (iii) < (ii) < (i)
- (ii) < (iii) < (i)

7. Which amongst the following reactions of alkyl halides produces isonitrile as a major product? (2023-Manipur)



Choose the **most appropriate** answer from the options given below:

- (D) only
- (C) and (D) only
- (B) only
- (A) and (B) only

8. The major product formed in dehydrohalogenation reaction of 2-Bromo pentane is Pent-2-ene. This product formation is based on? (2021)

- a. Hund's rule b. Hofmann rule
c. Huckel's rule d. Saytzeff's rule

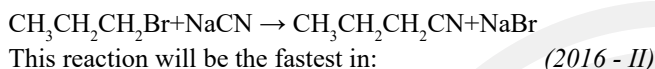
9. Elimination reaction of 2-Bromo-pentane to form pent-2-ene is (2020)

- (1) β -Elimination reaction
(2) Follows Zaitsev rule
(3) Dehydrohalogenation reaction
(4) Dehydration reaction
a. (1), (3), (4) b. (2), (3), (4)
c. (1), (2), (4) d. (1), (2), (3)

10. An example of a sigma bonded organometallic compound is: (2017-Delhi)

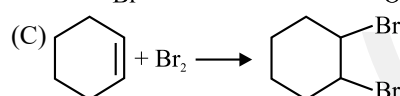
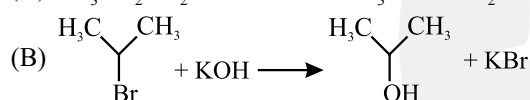
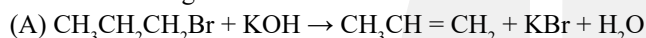
- a. Cobaltocene b. Ruthenocene
c. Grignard's d. Ferrocene

11. Consider the reaction



- a. N,N'-dimethylformamide (DMF)
b. Water
c. Ethanol
d. Methanol

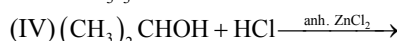
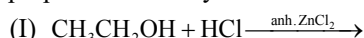
12. For the following reaction:



Which of the following statements is correct? (2016 - I)

- a. (A) is substitution, (B) and (C) are addition reactions.
b. (A) and (B) are elimination reactions and (C) is addition reaction.
c. (A) is elimination, (B) is substitution and (C) is addition reaction.
d. (A) is elimination, (B) and (C) are substitution reactions.

13. Which of the following reaction(s) can be used for the preparation of alkyl halides? (2015 Re)



- a. (III) and (IV) only b. (I), (III) and (IV) only
c. (I) and (II) only d. (IV) only

Stereochemical Aspects of Nucleophilic Substitution Reactions

14. Predict the order of reactivity of the following four isomers towards $\text{S}_\text{N}2$ reaction. (2022 Re)

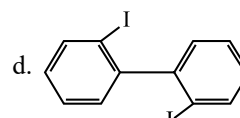
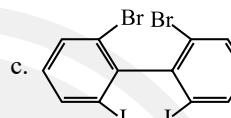
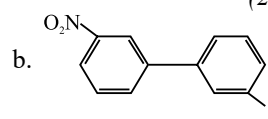
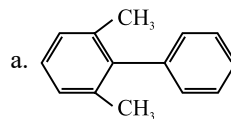
- (I) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$ (II) $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CH}_3$
(III) $(\text{CH}_3)_2\text{CHCH}_2\text{Cl}$ (IV) $(\text{CH}_3)_3\text{CCl}$

- a. (IV) > (II) > (III) > (I)
b. (IV) > (III) > (II) > (I)
c. (I) > (II) > (III) > (IV)
d. (I) > (III) > (II) > (IV)

15. The **incorrect** statement regarding chirality is: (2022)

- a. A racemic mixture shows zero optical rotation.
b. $\text{S}_\text{N}1$ reaction yields 1 : 1 mixture of both enantiomers.
c. The product obtained by $\text{S}_\text{N}2$ reaction of haloalkane having chirality at the reactive site shows inversion of configuration.
d. Enantiomers are superimposable mirror images on each other.

16. Which of the following biphenyls is optically active? (2016 - I)



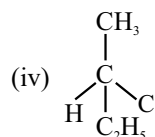
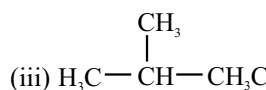
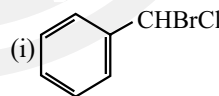
17. Two possible stereo-structures of $\text{CH}_3\text{CHOHCOOH}$, which are optically active, are called: (2015 Re)

- a. Mesomers b. Diastereomers
c. Atropisomers d. Enantiomers

18. In a $\text{S}_\text{N}1$ reaction on chiral centres, there is: (2015 Re)

- a. 100% inversion
b. 100% racemisation
c. Inversion more than retention leading to partial racemisation
d. 100% retention

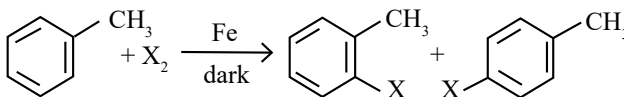
19. Which of the following compounds will undergo racemisation when solution of KOH hydrolyses? (2014)



- a. (ii) and (iv) b. (iii) and (iv)
c. (i) and (iv) d. (i) and (ii)

Haloarenes and Chemical Reactions of Haloarenes

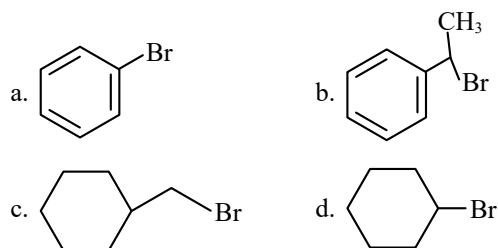
20. The following reaction method



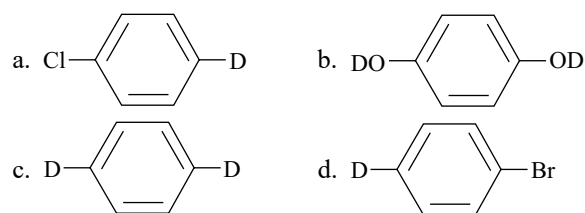
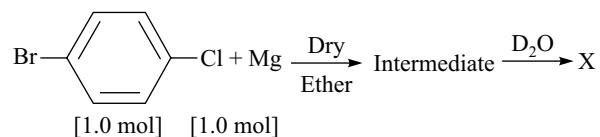
is not suitable for the preparation of the corresponding haloarene products, due to high reactivity of halogen, when X is: (2024 Re)

- a. F b. I c. Cl d. Br

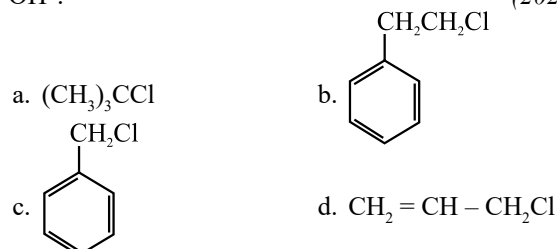
21. The compound that will undergo S_N1 reaction with the fastest rate is (2024)



22. Identify 'X' in the following reaction. (2023-Manipur)



23. Which of the following will NOT undergo S_N1 reaction with OH^- ? (2020-Covid)



Answer Key

1. (d) 2. (a) 3. (b) 4. (c) 5. (b) 6. (d) 7. (c) 8. (d) 9. (d) 10. (c)
 11. (a) 12. (c) 13. (b) 14. (d) 15. (d) 16. (c) 17. (d) 18. (c) 19. (c) 20. (a)
 21. (b) 22. (a) 23. (b)